Exercise 4.4

1. Find the square roots of the following polynomials by factorization method:

(i) $x^2 - 8x + 16$

$$x^2 - 8x + 16 = (x)^2 - 2(x)(4) + (4)^2$$
$$x^2 - 8x + 16 = (x - 4)^2$$

Taking square root on both sides

$$\sqrt{x^2 - 8x + 16} = \sqrt{(x-4)^2}$$
$$\sqrt{\boldsymbol{x^2 - 8x + 16}} = \pm(\boldsymbol{x - 4})$$

(ii) $9x^2 + 12x + 4$

$$9x^2 + 12x + 4 = (3x)^2 + 2(3x)(2) + (2)^2$$
$$9x^2 + 12x + 4 = (3x + 2)^2$$

Taking square root on both sides

$$\sqrt{9x^2 + 12x + 4} = \sqrt{(3x+2)^2}$$
$$\sqrt{\boldsymbol{9x^2 + 12x + 4}} = \pm(\boldsymbol{3x + 2})$$

(iii) $36a^2 + 84a + 49$

$$36a^2 + 84a + 49 = (6a)^2 + 2(6a)(7) + (7)^2$$
$$36a^2 + 84a + 49 = (6a + 7)^2$$

Taking square root on both sides

$$\sqrt{36a^2 + 84a + 49} = \sqrt{(6a+7)^2}$$
$$\sqrt{\boldsymbol{36a^2 + 84a + 49}} = \pm(\boldsymbol{6a + 7})$$

(iv) $64y^2 - 32y + 4$

$$64y^2 - 32y + 4 = (8y)^2 - 2(8y)(2) + (2)^2$$
$$64y^2 - 32y + 4 = (8y - 2)^2$$

Taking square root on both sides

$$\sqrt{64y^2 - 32y + 4} = \sqrt{(8y-2)^2}$$
$$\sqrt{\boldsymbol{64y^2 - 32y + 4}} = \pm(\boldsymbol{8y - 2})$$

(v) $200t^2 - 120t + 18$

$$200t^2 - 120t + 18 = 2[100t^2 - 60t + 9]$$
$$200t^2 - 120t + 18 = 2[(10t)^2 - 2(10t)(3) + (3)^2]$$
$$200t^2 - 120t + 18 = 2(8y - 2)^2$$

Taking square root on both sides

$$\sqrt{200t^2 - 120t + 18} = \sqrt{2(8y-2)^2}$$
$$\sqrt{\boldsymbol{200t^2 - 120t + 18}} = \pm\sqrt{\boldsymbol{2}}(\boldsymbol{8y - 2})$$

(vi) $40x^2 + 120x + 90$

$$40x^2 + 120x + 90 = 10[4x^2 + 12x + 9]$$
$$40x^2 + 120x + 90 = 10[(2x)^2 + 2(2x)(3) + (3)^2]$$
$$40x^2 + 120x + 90 = 10(2x + 3)^2$$

Taking square root on both sides

$$\sqrt{40x^2 + 120x + 90} = \sqrt{10(2x+3)^2}$$
$$\sqrt{\boldsymbol{40x^2 + 120x + 90}} = \pm\sqrt{\boldsymbol{10}}(\boldsymbol{2x + 3})$$

2. Find the square root of the following polynomials by division method:

(i) $4x^4 - 28x^3 + 37x^2 + 42x + 9$

$$
\begin{array}{r|l}
 & 2x^2 - 7x - 3 \\
\hline
2x^2 & 4x^4 - 28x^3 + 37x^2 + 42x + 9 \\
 & \pm 4x^4 \\
\hline
4x^2 - 7x & -28x^3 + 37x^2 + 42x + 9 \\
 & \mp 28x^3 \pm 49x^2 \\
\hline
4x^2 - 14x - 3 & -12x^2 + 42x + 9 \\
 & \mp 12x^2 \pm 42x \pm 9 \\
\hline
 & 0
\end{array}
$$

$$\boldsymbol{Square\ Root = \pm(2x^2 - 7x - 3)}$$

(ii) $121x^4 - 198x^3 - 183x^2 + 216x + 144$

$$
\begin{array}{r|l}
 & 11x^2 - 9x - 12 \\
\hline
11x^2 & 121x^4 - 198x^3 - 183x^2 + 216x + 144 \\
 & \pm 121x^4 \\
\hline
22x^2 - 9x & -198x^3 - 183x^2 + 216x + 144 \\
 & \mp 198x^3 \pm 81x^2 \\
\hline
22x^2 - 18x - 12 & -264x^2 + 216x + 144 \\
 & \mp 264x^2 \pm 216x \pm 144 \\
\hline
 & 0
\end{array}
$$

$$\boldsymbol{Square\ Root = \pm(11x^2 - 9x - 12)}$$

(iii) $x^4 - 10x^3y + 27x^2y^2 - 10xy^3 + y^4$

$$
\begin{array}{r|l}
 & x^2 - 5xy + y^2 \\
\hline
x^2 & x^4 - 10x^3y + 27x^2y^2 - 10xy^3 + y^4 \\
 & \pm x^4 \\
\hline
2x^2 - 5xy & -10x^3y + 27x^2y^2 - 10xy^3 + y^4 \\
 & \mp 10x^3y \pm 25x^2y^2 \\
\hline
2x^2 - 10xy + y^2 & 2x^2y^2 - 10xy^3 + y^4 \\
 & \pm 2x^2y^2 \mp 10xy^3 \pm y^4 \\
\hline
 & 0
\end{array}
$$

$$\boldsymbol{Square\ Root = \pm(x^2 - 5xy + y^2)}$$

(iv) $4x^4 - 12x^3 + 37x^2 - 42x + 49$

$$
\begin{array}{r|l}
 & 2x^2 - 3x + 7 \\
\hline
2x^2 & 4x^4 - 12x^3 + 37x^2 - 42x + 49 \\
 & \pm 4x^4 \\
\hline
4x^2 - 3x & -12x^3 + 37x^2 - 42x + 49 \\
 & \mp 12x^3 \pm 9x^2 \\
\hline
4x^2 - 6x + 7 & 28x^2 - 42x + 49 \\
 & \pm 12x^2 \mp 42x \pm 49 \\
\hline
 & 0
\end{array}
$$

$$\boldsymbol{Square\ Root = \pm(2x^2 - 3x - 7)}$$

3. An investor's return $R(x)$ in rupees after investing x thousand rupees is given by quadratic expression: $R(x) = -x^2 + 6x - 8$ Factorize the expression and find the investment levels that result in zero return.

$$R(x) = -x^2 + 6x - 8$$
$$R(x) = -[x^2 - 6x + 8]$$
$$R(x) = -[x^2 - 4x - 2x + 8]$$
$$R(x) = -[x(x-4) - 2(x-4)]$$
$$R(x) = -[(x-2)(x-4)]$$
$$R(x) = -(x-2)(x-4)$$

Put $R(x) = 0$ for zero return:

$$R(x) = 0$$
$$-(x-2)(x-4) = 0$$

Either $\quad x - 2 = 0 \quad$ or $\quad x - 4 = 0$

$x = 2 \quad , \quad x = 4$

Since $\boldsymbol{x}$ is in thousands of rupees, the investment levels that result in zero return are $\boldsymbol{Rs. 2000}$ and $\boldsymbol{Rs. 4000}$.

4. A company's profit $\boldsymbol{P(x)}$ in rupees from selling $\boldsymbol{x}$ units of a product is modeled by the cubic expression: $\boldsymbol{P(x) = x^3 - 15x^2 + 75x - 125}$ Find the break-even point(s), where the profit is zero.

$$P(x) = x^3 - 15x^2 + 75x - 125$$
$$P(x) = (x)^3 - 3(x)^2(5) + 3(x)(5)^2 - (5)^3$$
$$P(x) = (x-5)^3$$

Put $P(x) = 0$ for break-even point(s)

$$P(x) = 0$$
$$(x-5)^3 = 0$$

Taking cube root on both sides

$$\sqrt[3]{(x-5)^3} = \sqrt[3]{0}$$
$$x - 5 = 0$$
$$x = 5$$

Thus, the break-even point is at **5 units**.

5. The potential energy $\boldsymbol{V(x)}$ in an electric field varies as a cubic function of distance $\boldsymbol{x}$, given by: $\boldsymbol{V(x) = 2x^3 - 6x^2 + 4x}$ Determine where the potential energy is zero.

$$V(x) = 2x^3 - 6x^2 + 4x$$
$$V(x) = 2x[x^2 - 3x + 2]$$
$$V(x) = 2x[x^2 - 2x - x + 2]$$
$$V(x) = 2x[x(x-2) - 1(x-2)]$$
$$V(x) = 2x[(x-2)(x-1)]$$
$$V(x) = 2x(x-2)(x-1)$$

Put $V(x) = 0$ for zero potential energy

$$V(x) = 0$$
$$2x(x-2)(x-1) = 0$$

$\Longrightarrow \quad 2x = 0 \quad , \quad x - 2 = 0 \quad , \quad x - 1 = 0$

$x = 0 \quad , \quad x = 2 \quad , \quad x = 1$

Thus, the potential energy is zero at $\boldsymbol{x = 0}$, $\boldsymbol{x = 1}$, and $\boldsymbol{x = 2}$.

6. In structural engineering, the deflection $\boldsymbol{Y(x)}$ of a beam is given by: $\boldsymbol{Y(x) = 2x^2 - 8x + 6}$. This equation gives the vertical deflection at any point $\boldsymbol{x}$ along the beam. Find the points of zero deflection.

$$Y(x) = 2x^2 - 8x + 6$$
$$Y(x) = 2[x^2 - 4x + 3]$$
$$Y(x) = 2[x^2 - 3x - x + 3]$$
$$Y(x) = 2[x(x-3) - 1(x-3)]$$
$$Y(x) = 2[(x-3)(x-1)]$$
$$Y(x) = 2(x-3)(x-1)$$

Put $Y(x) = 0$ for points of zero deflection

$$Y(x) = 0$$

$$2(x-3)(x-1)=0$$

Since $2 \neq 0$, so

Either $x-3=0$ or $x-1=0$

$x=3$, $x=1$

Therefore, the points of zero deflection are at $\boldsymbol{x = 1}$ and $\boldsymbol{x = 3}$.